

Software Requirements Specification (SRS)

Preface

This document defines the software requirements for the Asset & Resource Allocation Management System. It is intended for students, instructors, project reviewers, developers, and testers.

Version History

Version	Date	Description
1.0	Initial	First version of the Software Requirements Specification created for academic submission.

Rationale: This document has been created as part of the Requirement Engineering process to formally specify system requirements.

1. Introduction

1.1 Need for the System

Organizations often rely on manual or fragmented methods to manage assets and resources. This results in inefficient utilization, lack of transparency, delayed approvals, and difficulty in tracking maintenance and allocation history.

1.2 System Overview

The Asset & Resource Allocation Management System is a centralized web-based application that manages asset tracking, allocation requests, approval workflows, and maintenance activities.

1.3 System Functions (Brief)

The Resource Management System provides the following core system functions to manage institutional resources.

3.1 Resource Request Management

The system allows requesters to submit resource requests, track request status, and view request history.

3.2 Request Approval Management

The system allows department heads to review incoming requests and approve or reject them.

3.3 Resource Allocation Management

The system allows staff to allocate resources to approved requests and notify requesters after allocation.

3.4 Resource Inventory Management

The system maintains a centralized inventory of all resources and allows staff to add or update resource information.

3.5 Procurement Management

The system supports procurement requests for additional resources and allows registrars to approve or reject procurement requests.

3.6 Maintenance Management

The system allows requesters to report maintenance issues and allows staff to repair or scrap resources.

3.7 Resource Return Management

The system allows requesters to return allocated resources and enables staff to inspect and update the resource condition.

3.8 Analytics and Monitoring

The system provides dashboards and analytics to monitor resource usage and request trends.

3.9 User Management

The system allows administrators to create, edit, and manage system users.

3.10 Department Management

The system allows administrators to manage departments and assign department heads.

3.11 System Logs Management

The system records system activities and allows administrators to view and monitor system logs.

1.4 Interaction with Other Systems

The system is designed to work as a standalone application but may integrate with authentication or inventory systems in the future.

1.5 Business and Strategic Objectives

Improve asset utilization

Increase transparency and accountability

Reduce operational delays

Support informed decision-making

2. Glossary

Domain specific definitions

Term	Definition
Asset	A physical resource such as computers, equipment, or rooms
Allocation	Assignment of an asset to a user or office
Register	Senior authority responsible for procurement approvals
Audit Log	Record of system activities for accountability

3. User Requirements Definition

This section describes the services provided to system users. Requirements are expressed in natural language and are understandable to non-technical stakeholders.

3.1 Functional Requirements

The functional requirements describe the operations and services that the Resource Management System must perform.

3.1.1 Resource Request Module

- FR-1 The system shall allow requesters to submit resource requests including department, resource type, quantity, and reason.
- FR-2 The system shall generate a unique request ID for each request.
- FR-3 The system shall allow requesters to request a new resource type if the resource does not exist.
- FR-4 The system shall forward submitted requests to the department head.
- FR-5 The system shall allow requesters to view all submitted requests.
- FR-6 The system shall display request status including Pending, Approved, Rejected, and Allocated.
- FR-7 The system shall allow requesters to confirm resource receipt after allocation.
- FR-8 The system shall maintain history of completed and rejected requests.

3.1.2 Resource Allocation Module

- FR-9 The system shall allow staff to view approved requests awaiting allocation.
- FR-10 The system shall allow staff to assign resource codes to approved requests.
- FR-11 The system shall record allocation details including request ID and resource code.
- FR-12 The system shall notify requesters when resources are allocated.
- FR-13 The system shall maintain allocation history records.

3.1.3 Resource Management Module

- FR-14 The system shall allow staff and department heads to view resource inventory.
- FR-15 The system shall allow staff to add new resources to the system.
- FR-16 The system shall allow staff to update resource information.
- FR-17 The system shall notify the department head when resource details are modified.
- FR-18 The system shall track resource status including Available, Allocated, Maintenance, and Scrap.
- FR-19 The system shall maintain resource lifecycle history.

3.1.4 Procurement Management Module

- FR-20 The system shall allow department heads to create procurement requests.
- FR-21 The system shall allow the registrar to view procurement requests.
- FR-22 The system shall allow the registrar to approve procurement requests.
- FR-23 The system shall allow the registrar to reject procurement requests.
- FR-24 The system shall create procurement tasks for staff after approval.
- FR-25 The system shall allow staff to register newly purchased resources.
- FR-26 The system shall store purchase information including vendor details and warranty.
- FR-27 The system shall maintain procurement request history.

3.1.5 Maintenance Management Module

- FR-28 The system shall allow requesters to submit maintenance requests for allocated

resources.

FR-29 The system shall allow staff to review maintenance requests.

FR-30 The system shall allow staff to mark resources as repaired.

FR-31 The system shall allow staff to mark resources as scrap if not repairable.

FR-32 The system shall notify requesters about maintenance outcomes.

FR-33 The system shall maintain maintenance history records.

3.1.6 Resource Return Module

FR-34 The system shall allow requesters to submit return requests for allocated resources.

FR-35 The system shall allow staff to inspect returned resources.

FR-36 The system shall allow staff to mark returned resources as available or scrap.

FR-37 The system shall maintain return history records.

3.1.7 Analytics and Monitoring Module

FR-38 The system shall allow department heads to view department resource analytics.

FR-39 The system shall allow the registrar to view system-wide analytics.

FR-40 The system shall notify department heads when resource stock falls below the threshold.

3.1.8 User Management Module

FR-41 The system shall allow administrators to create user accounts.

FR-42 The system shall allow administrators to edit user details.

FR-43 The system shall allow administrators to assign roles to users.

FR-44 The system shall allow administrators to assign departments to users.

3.1.9 Department Management Module

FR-45 The system shall allow administrators to create departments.

FR-46 The system shall allow administrators to edit department information.

FR-47 The system shall allow administrators to assign department heads.

3.1.10 System Logs Module

FR-48 The system shall record system activities such as requests, allocations, maintenance actions, and user updates.

FR-49 The system shall allow administrators to view system logs.

FR-50 The system shall allow administrators to filter logs by user, role, or date.

FR-51 The system shall allow administrators to search logs.

FR-52 The system shall allow administrators to export system logs.

3.2 Non-Functional Requirements

The non-functional requirements describe the quality attributes of the Resource Management System.

3.2.1 Performance

The system should respond to user actions within two seconds.

3.2.2 Security

The system shall implement role-based access control to restrict access to authorized users.

3.2.3 Scalability

The system should support increasing numbers of users and resources without performance degradation.

3.2.4 Reliability

The system should maintain accurate resource records and prevent duplicate entries.

3.2.5 Usability

The system should provide a clear and user-friendly dashboard interface.

3.2.6 Maintainability

The system should use modular architecture to simplify maintenance and updates.

3.2.7 Availability

The system should ensure high system availability with minimal downtime.

3.2.8 Compatibility

The system should support modern web browsers and responsive layouts.

4. System Architecture

The system follows a web-based client–server architecture:

Presentation Layer: Web interface for all actors.

Application Layer: Business logic, workflows, and access control.

Data Layer: Database for users, assets, requests, and audit logs.

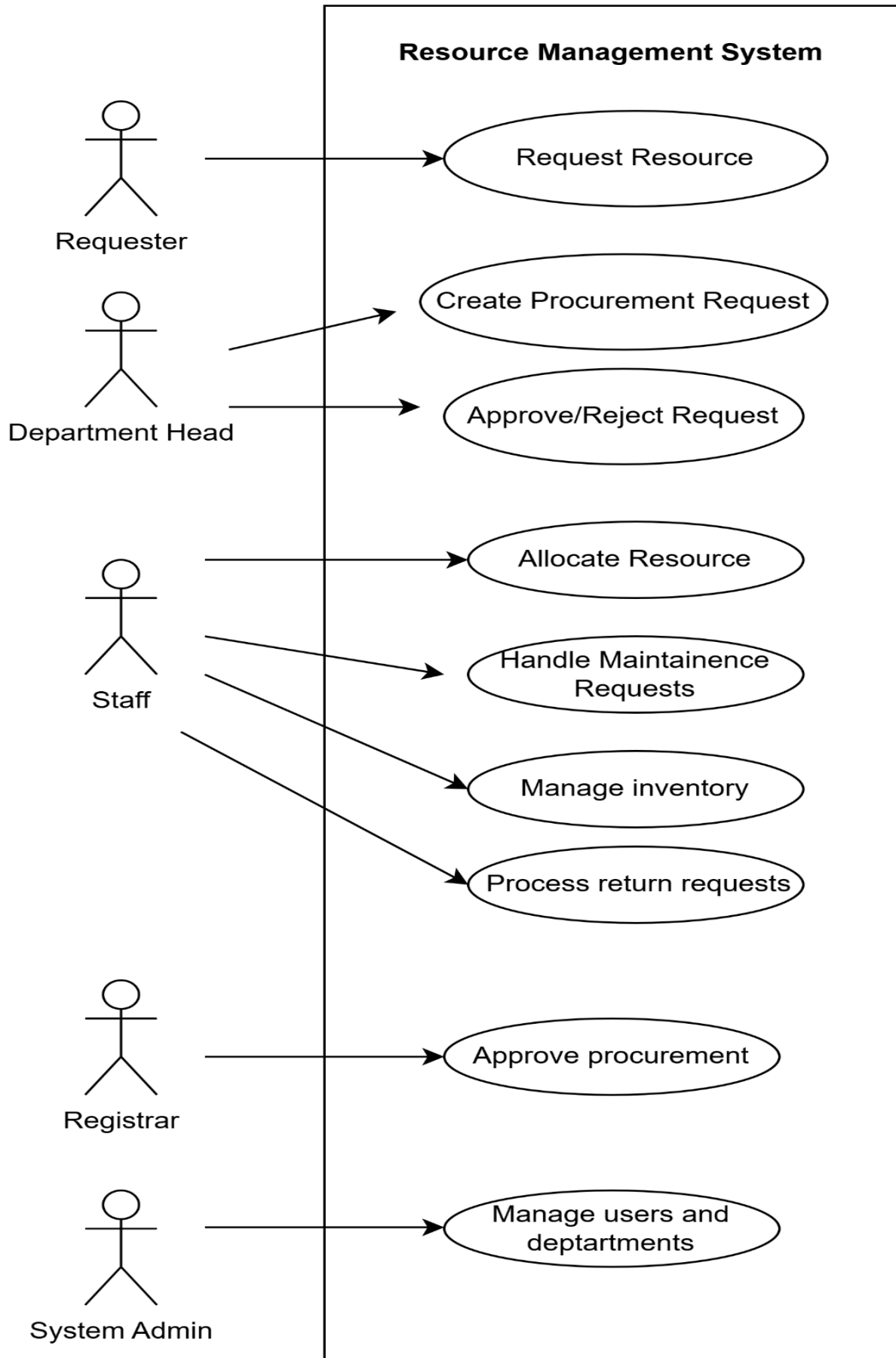
Reusable components include authentication and role-based authorization modules.

5. System Requirements Specification

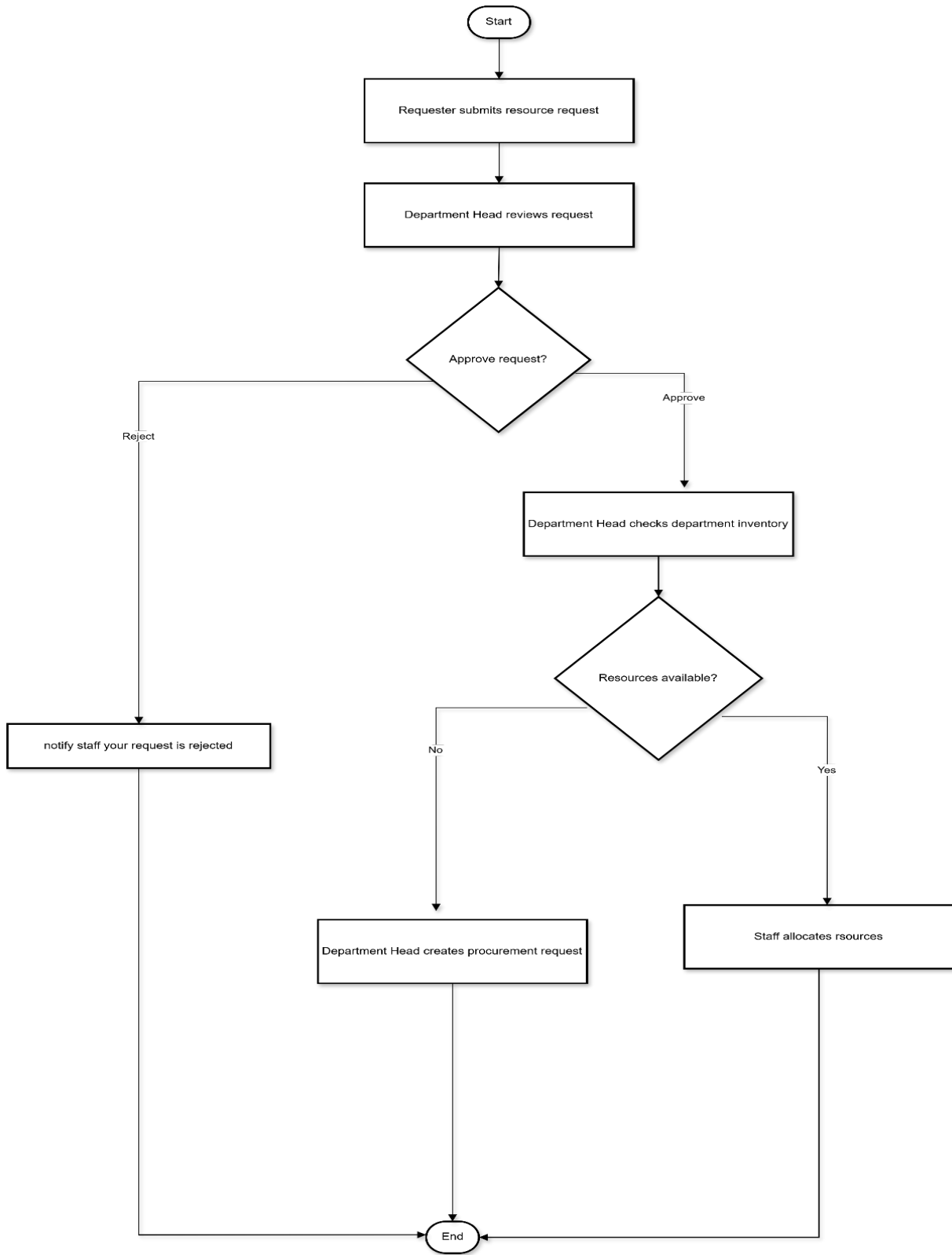
6. System Models

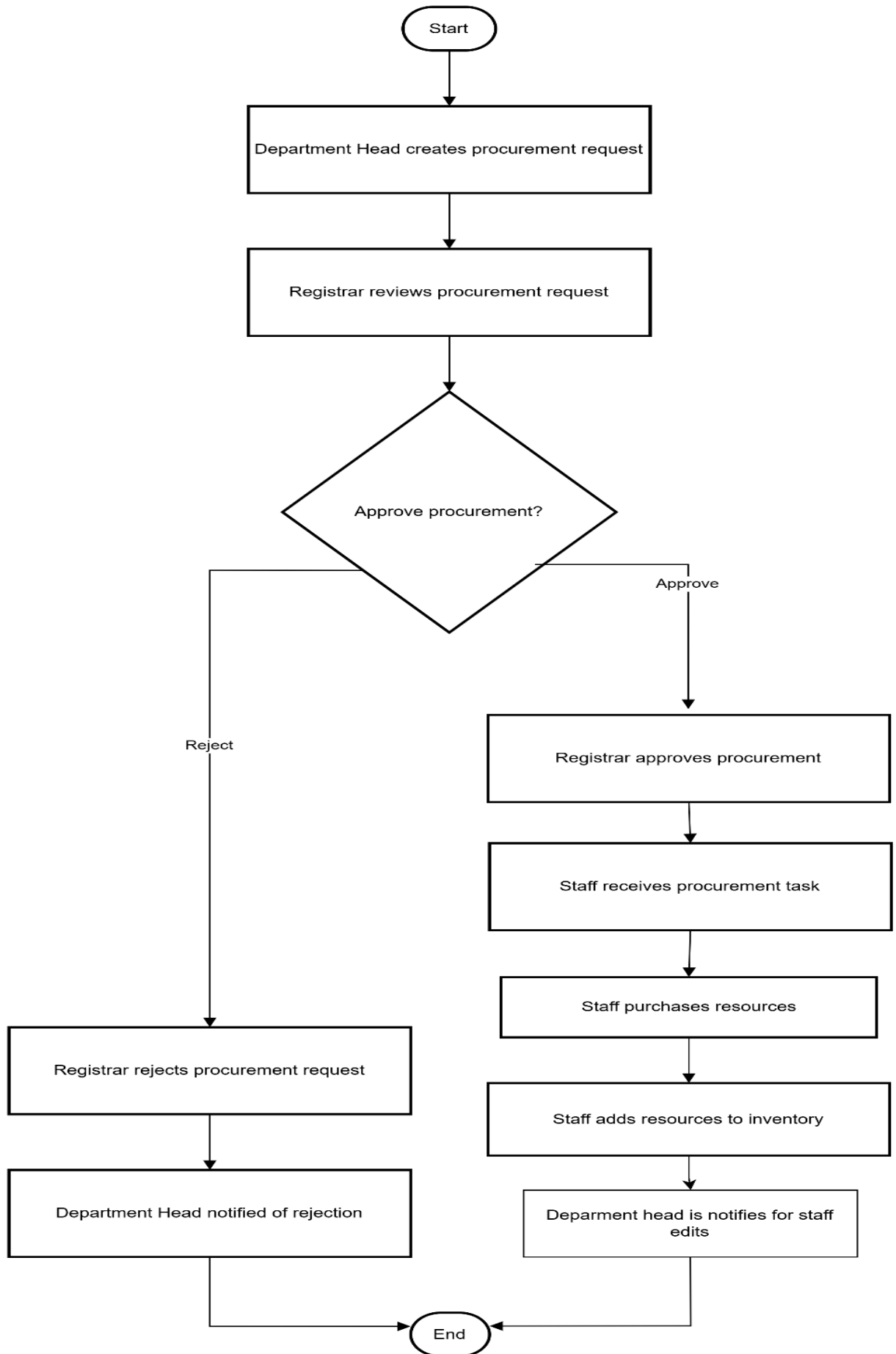
The following system models may be used:

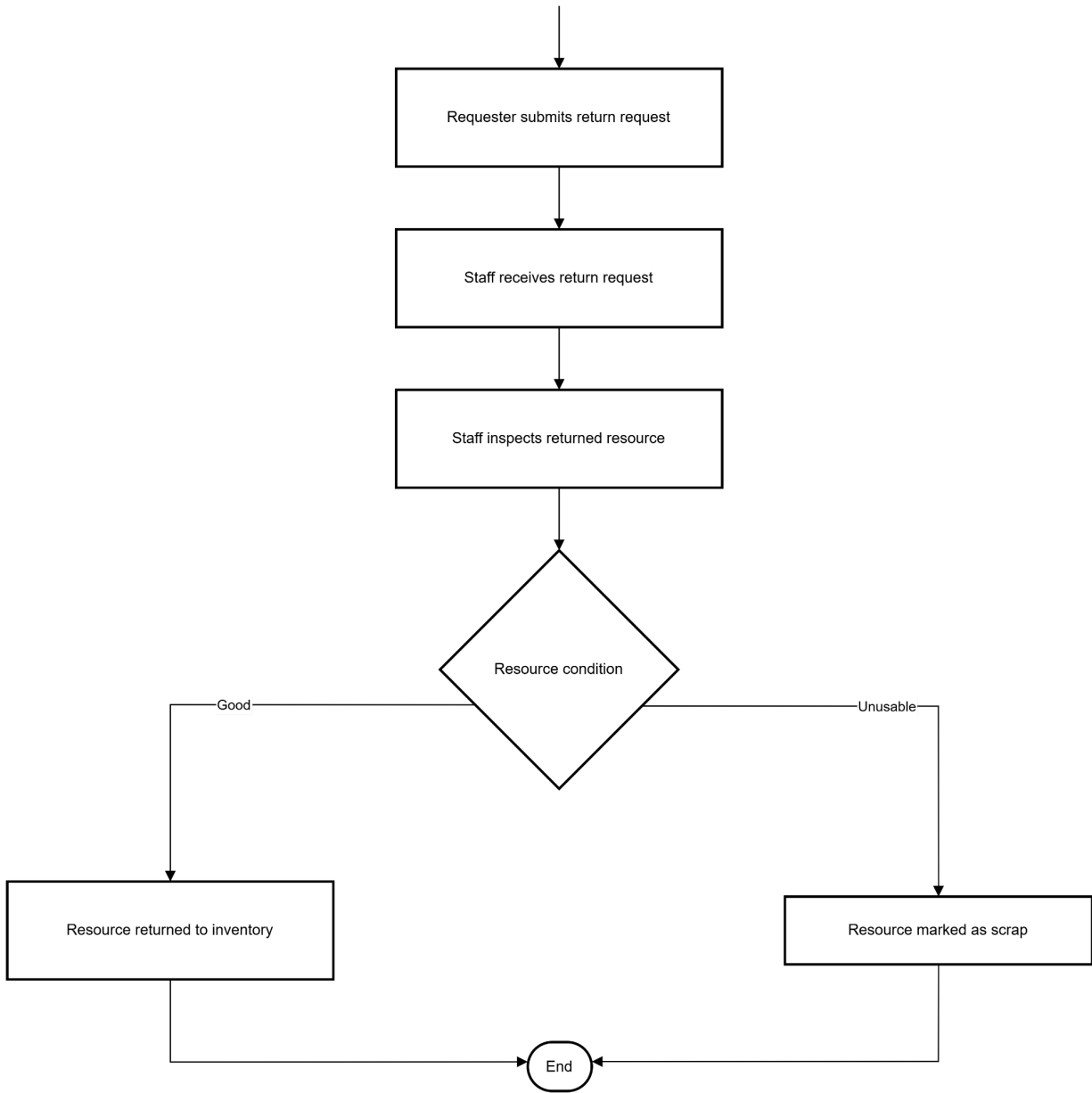
Use Case Diagram

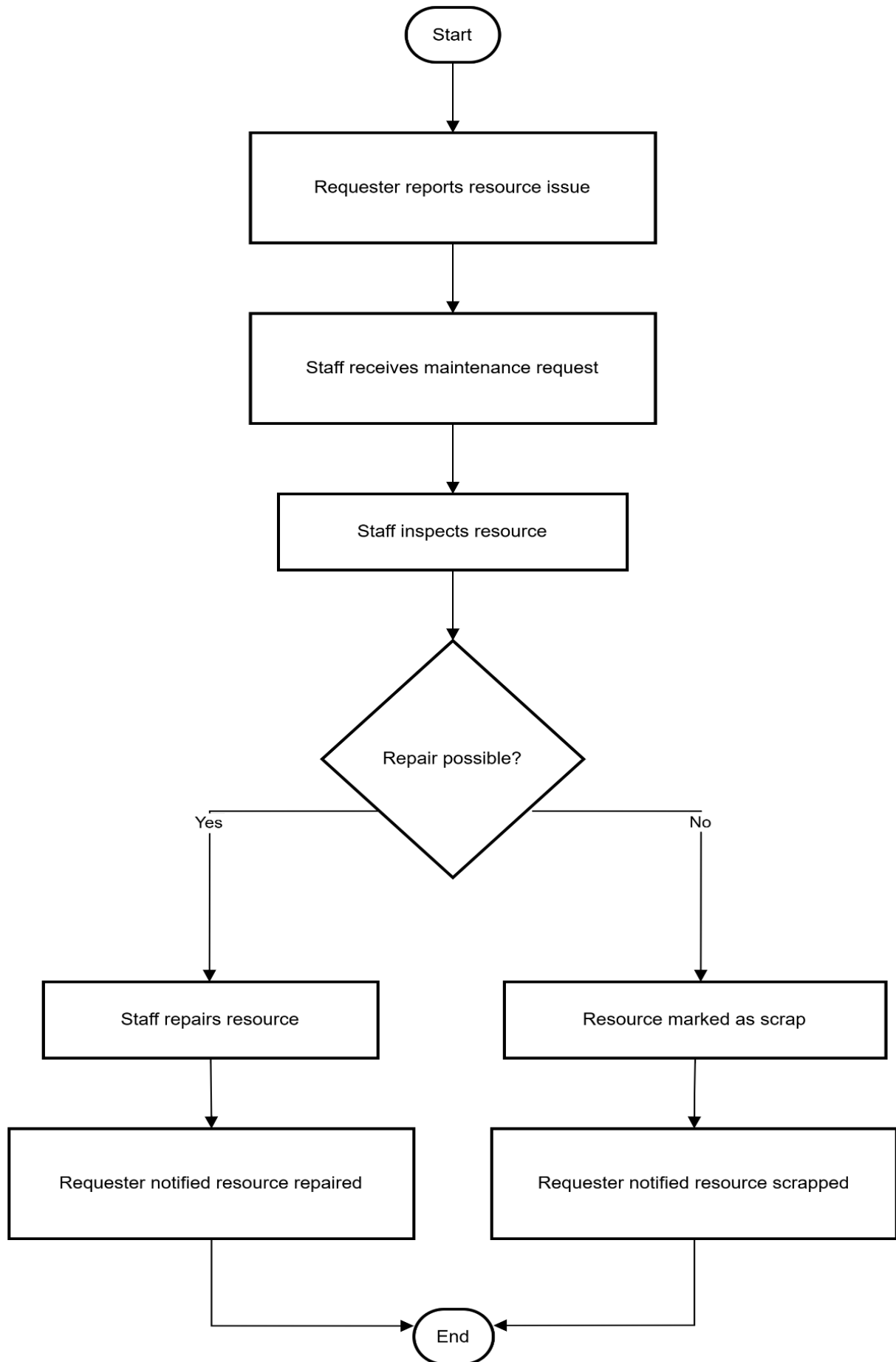


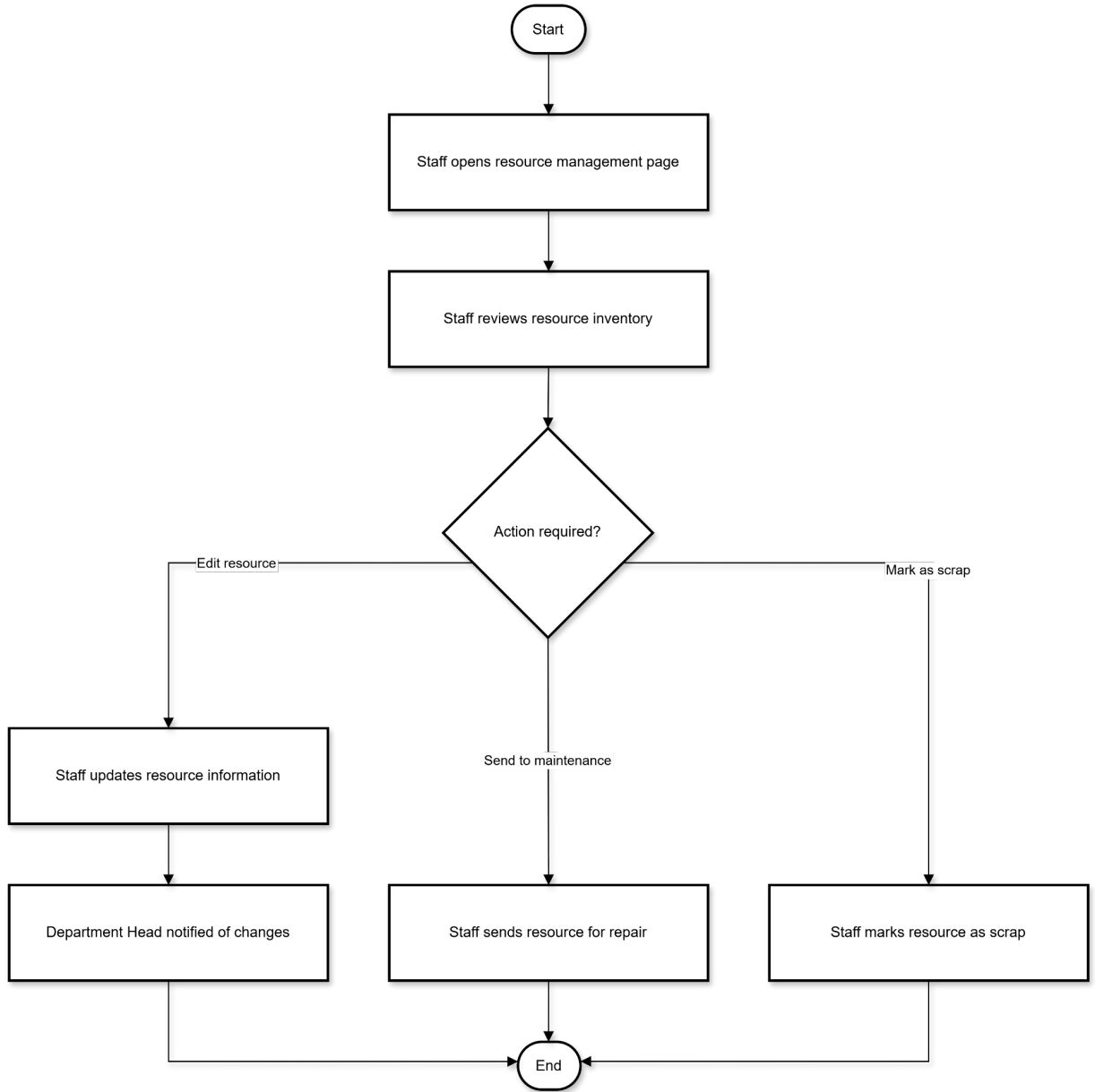
Activity Diagrams



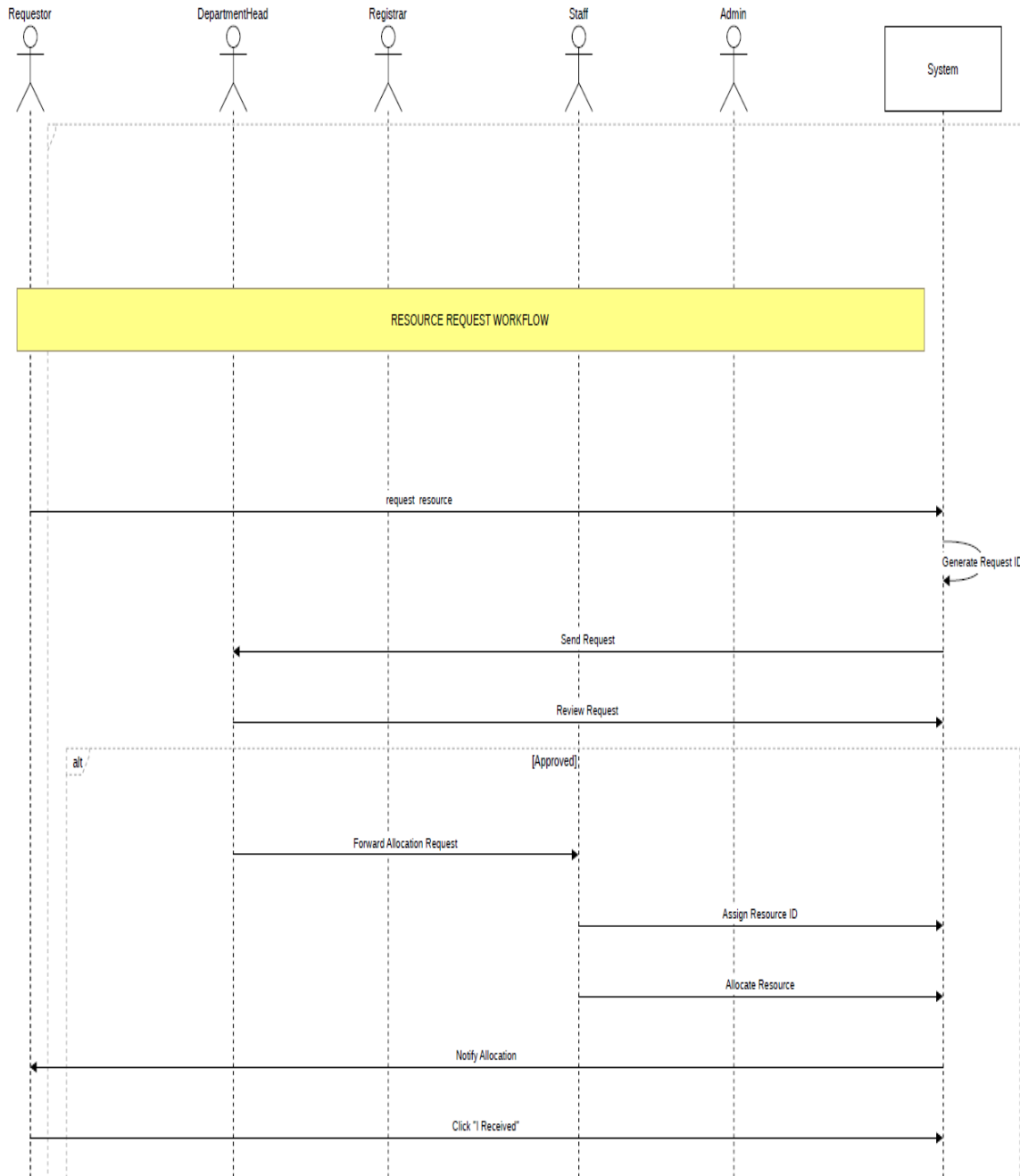


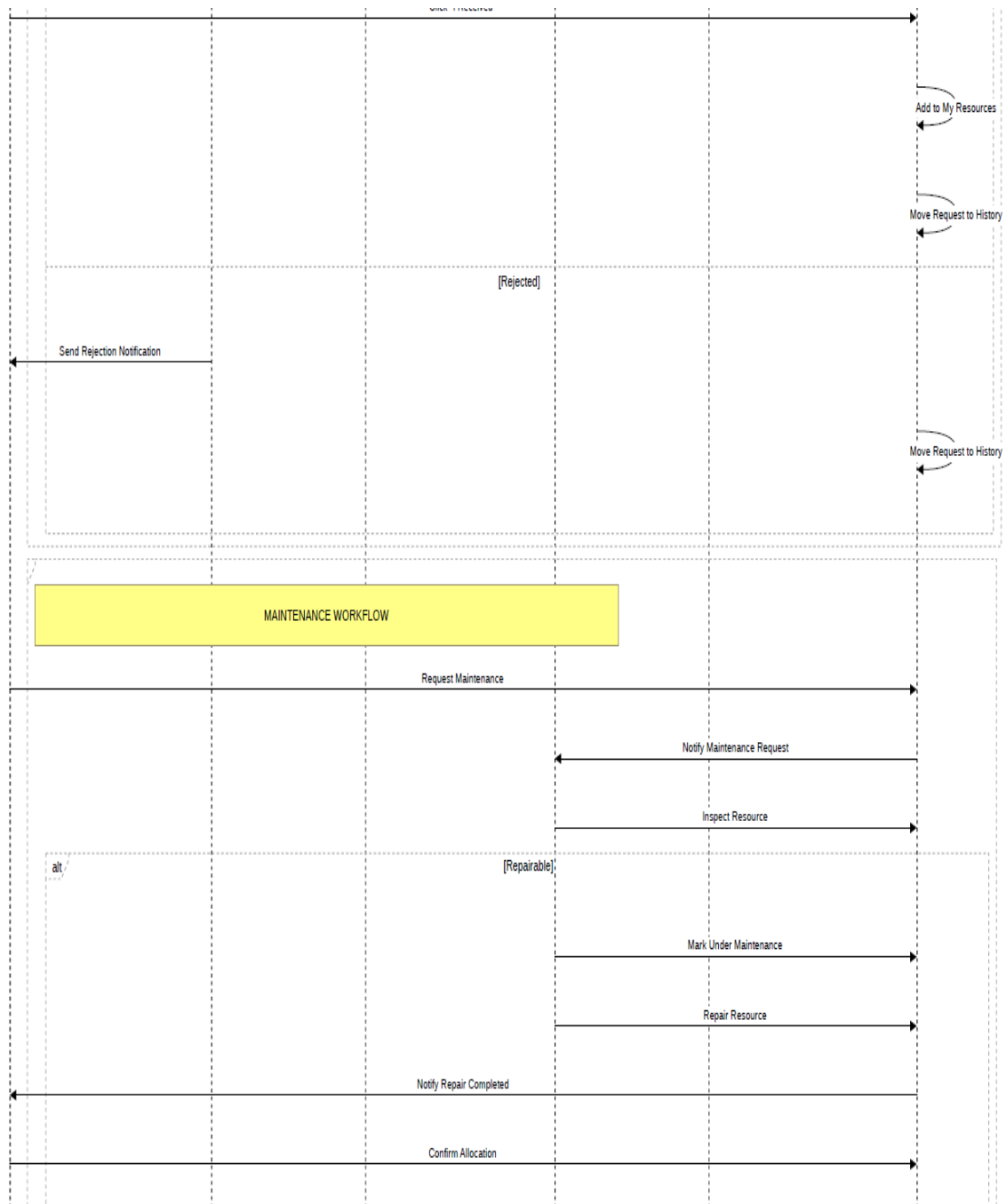


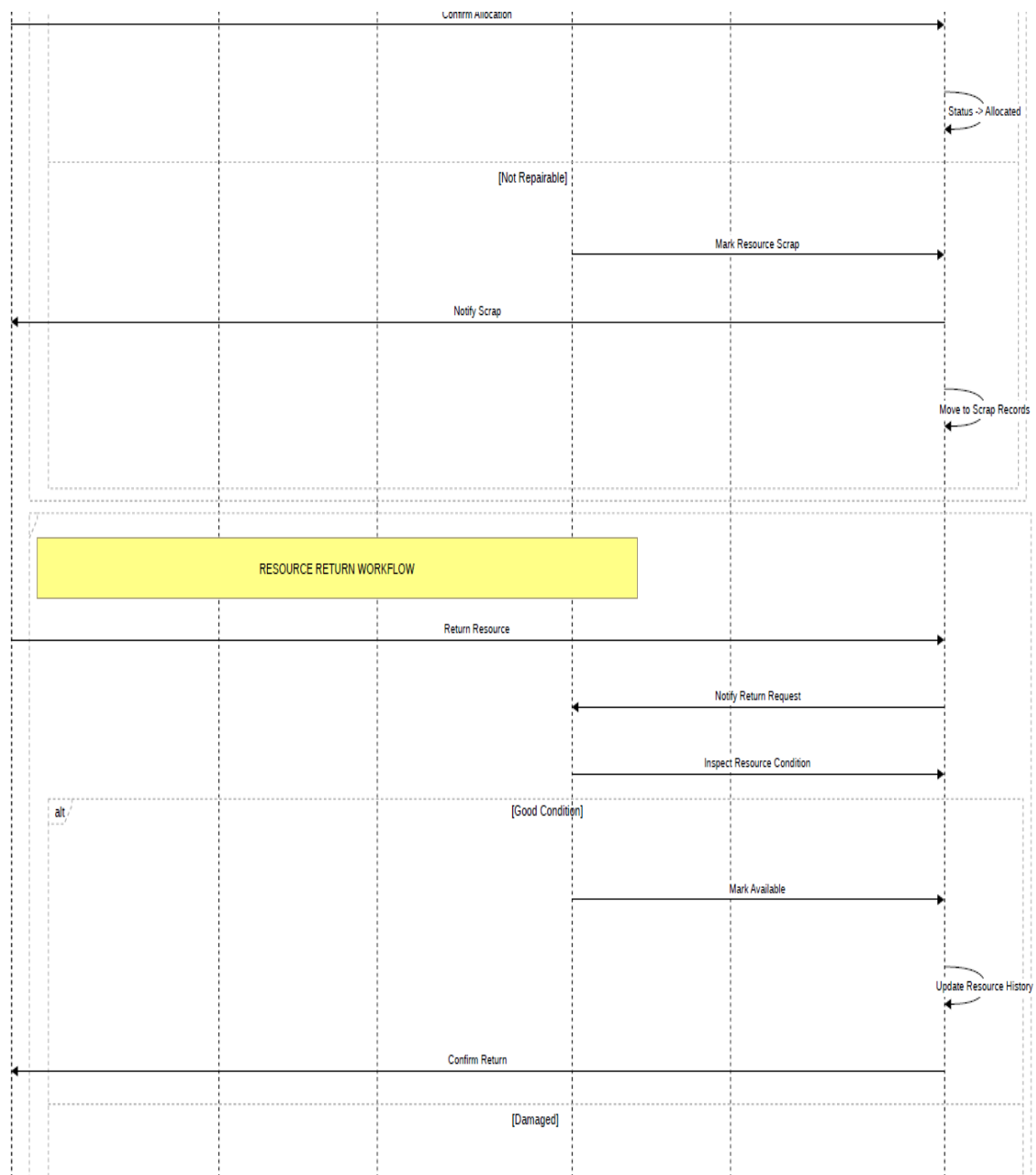


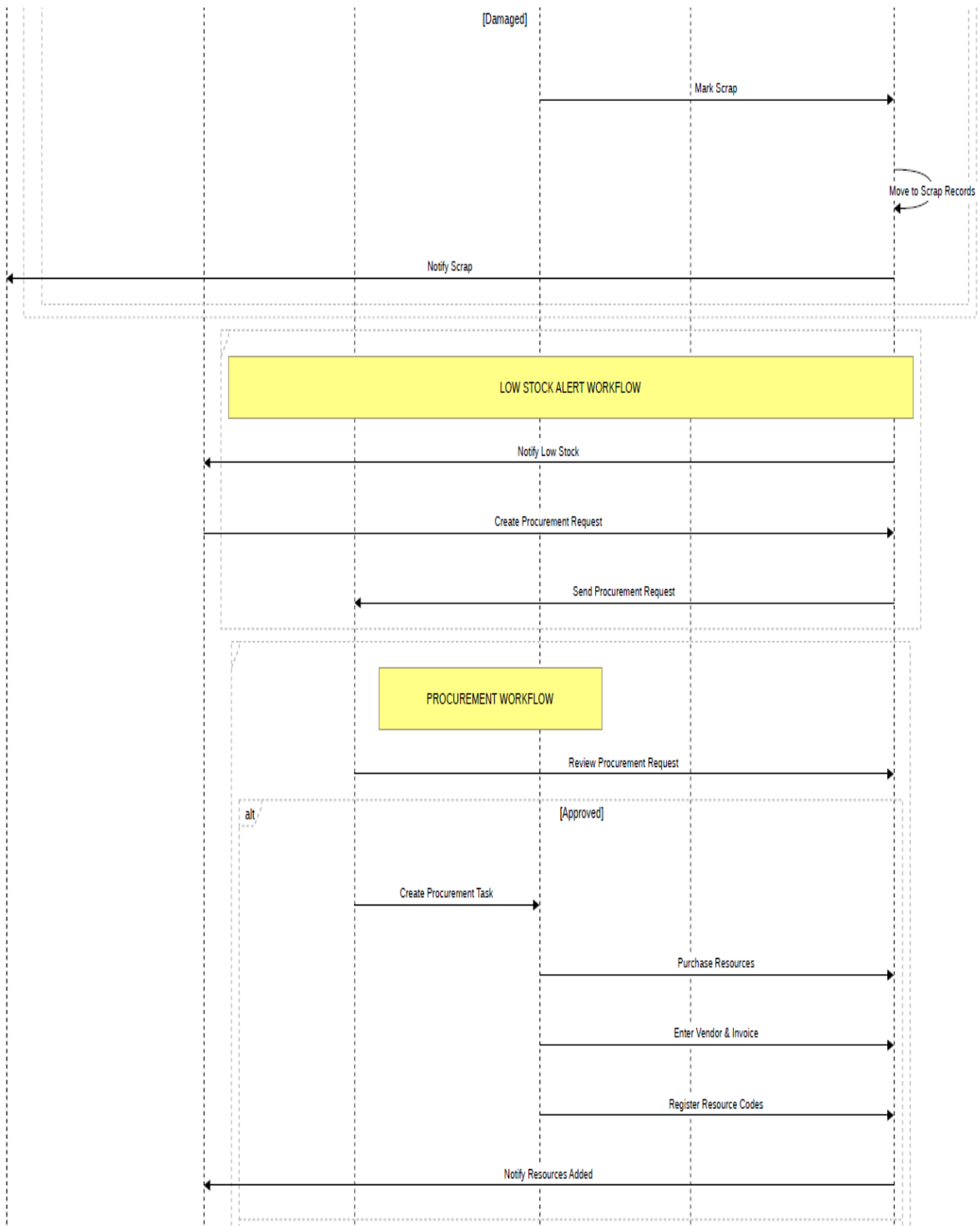


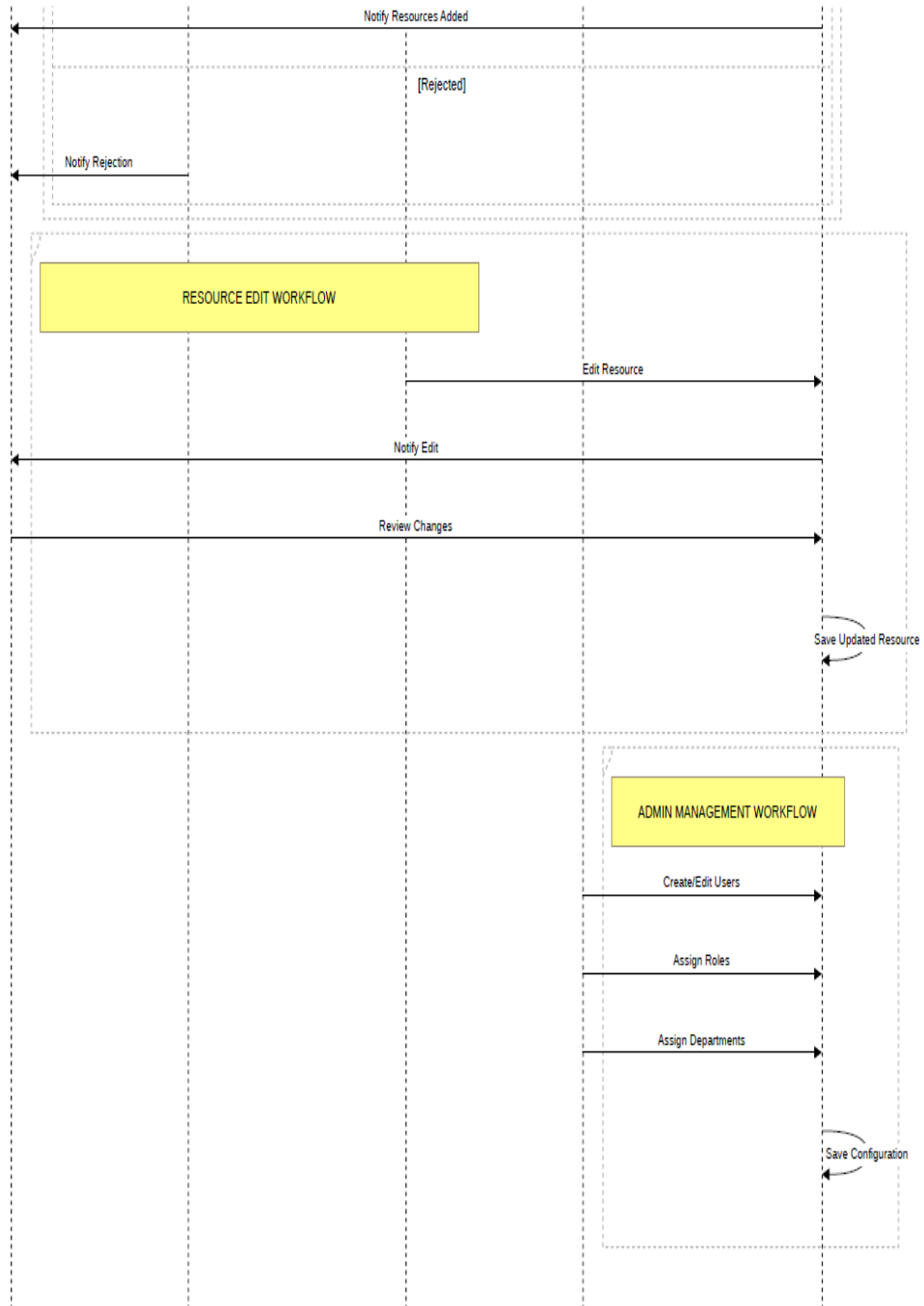
Sequence diagram



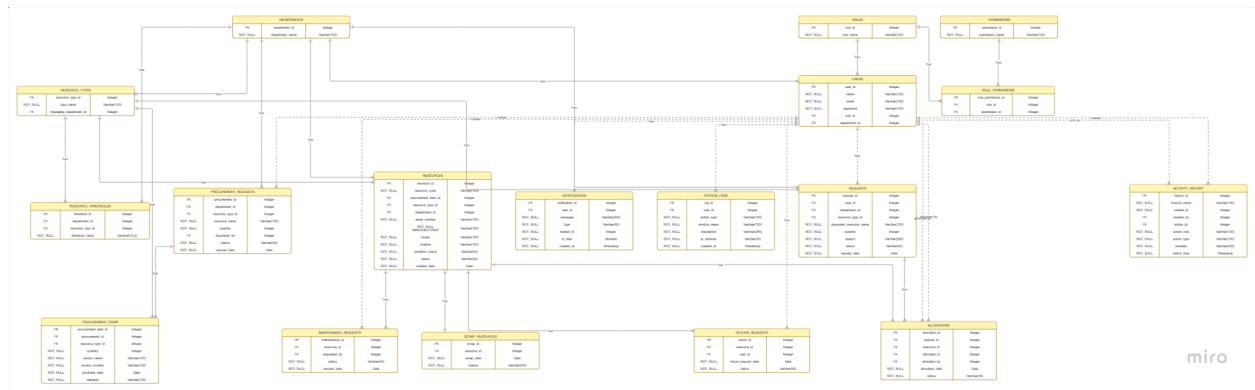








Entity Relationship (ER) Diagrams



7. System Evolution

The system is designed to support future enhancements such as mobile access, advanced analytics, and integration with external systems.

8. Appendices

8.1 Hardware Requirements

Server with minimum 8 GB RAM

Client systems with modern web browsers

8.2 Software Requirements

Web browser (Chrome, Firefox, Edge)

Backend server and database system

9. Index

Actors

Functional Requirements

Non-Functional Requirements

System Models